

2.1 PASSIVE DESIGN AND THERMAL SAFETY

Passive buildings are designed so that windows, walls, and floors are able to collect, store, and distribute solar energy in the form of heat in winter and reject solar heat in the summer. A passively designed house reduces the need for the use of mechanical and electrical (active heating and cooling) systems, saving energy and costs. For more information on passive design refer to: <http://www.yourhome.gov.au/passive-design>

With global warming temperatures predicted to increase a minimum of 1.5 degrees by 2030, Waverley Council is working to ensure that all new homes are built to be thermally safe to live and work in over the lifetime of the building.

Objectives

- (a) To encourage passive design to be integrated into every development from the design stage.
- (b) To encourage passive design through site layout, design and construction to reduce the need for active heating and cooling systems and electric lighting.
- (c) To ensure that local housing responds to regional climate conditions and remains thermally safe for occupants for the lifetime of the building:
 - a. as the climate warms
 - b. during the event of a power failure
- (d) To reduce the energy used in buildings.
- (e) To reduce peak electricity demand of developments.

Controls

- (a) Development is to be designed and constructed to incorporate passive design measures through site design and analysis. Refer to the Design Guidance for methods to achieve this.
- (b) Development must reduce solar heat gain with the following measures:
 - (i) Glazing on buildings must be high-performance low solar gain low-emissivity glass (single or double glazed units).
 - (ii) Skylights must be high-performance low-emissivity glass or double-glazed glass and should be ventilated.
- (c) Development must enable natural ventilation:
 - (i) Windows must be openable excluding windows that are for light ingress or privacy purposes.
 - (ii) Ceiling or wall mounted fans should be in all habitable rooms (main living areas and bedrooms). This should be notated on DA and CC plans.
- (d) Finishes must provide solar absorptance to mitigate the buildup of urban heat:
 - (i) Wall and roof finishes are to have a solar absorptance of < 0.475
 - (ii) Terracotta roofs are to have a solar absorptance of < 0.70
- (e) Development is to incorporate landscaping that provides canopy and vegetation for cooling to provide resilience during hot and dry periods.